Product Requirements Document (PRD)

RentiHub

Smart Commercial Rent & Property Management System

Version: 1.0

Prepared By: Group Project – Enterprise Architecture

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1. Introduction

1.1 Background

Many commercial buildings in Uganda still manage rent collection using physical books. Whenever a tenant pays rent, the payment is written in a book and both the tenant and rent collector sign.

Although this method has worked for years, it has many limitations:

- Payment records can easily be lost.
- Books can be damaged or stolen.
- There is no backup of payment information.
- Building owners cannot monitor rent collection in real time.
- It is difficult to know which tenants have paid or defaulted.
- Preparing financial reports is time-consuming.

RentiHub is designed to replace this manual process with a centralized digital platform that enables efficient rent collection, tenant management, communication, and reporting.

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2. Problem Statement

Commercial property owners and rent collectors face challenges in managing rent due to reliance on manual record keeping.

The absence of centralized digital records leads to:

- Lost payment history
- Human errors
- Payment disputes
- Poor accountability

- Delayed reporting
- Lack of transparency

The enterprise requires a secure and scalable solution that digitizes rent management while improving collaboration between all stakeholders.

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3. Purpose of the System

The purpose of RentiHub is to digitize commercial property management by providing a centralized platform for managing tenants, shops, rent payments, receipts, notifications, and reporting.

The system will improve operational efficiency while eliminating manual paperwork.

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4. Scope

The first version of the system will support:

- Commercial buildings
- Multiple floors
- Multiple shops
- Tenant management
- Rent collection
- Receipt generation
- Notifications
- Financial reports
- Maintenance requests
- Dashboard analytics

Future versions may include:

- Mobile Money integration
- Online payments
- QR Code receipts
- AI-based rent predictions
- Multi-company support

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5. Target Users

Building Owner

Responsibilities

- View reports
- Monitor income
- Monitor buildings
- View occupancy
- View unpaid rent

Property Manager / Rent Collector

Responsibilities

- Register tenants
- Record rent
- Generate receipts
- Manage shops
- View balances

Tenant

Responsibilities

- View payment history
- Download receipts
- Receive notifications
- Submit maintenance requests

Administrator

Responsibilities

- Manage users
- Manage permissions
- Configure system settings
- View audit logs

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6. Functional Requirements

Authentication

The system shall:

- Register users
- Login users
- Reset passwords
- Support role-based access

Roles include:

- Admin
- Owner
- Property Manager
- Tenant

Building Management

Users shall be able to:

- Add buildings
- Edit buildings
- Delete buildings
- View buildings

Each building contains:

- Name
- Address
- Floors

Floor Management

The system shall allow:

- Create floor
- Edit floor
- Delete floor

Example

Ground Floor

First Floor

Second Floor

Shop Management

Each shop shall contain:

- Shop Number
- Floor
- Monthly Rent
- Status

Status

- Occupied
- Vacant
- Reserved

Tenant Management

Store

- Full Name
- National ID
- Phone Number
- Email
- Business Name
- Shop Assigned
- Lease Start Date
- Lease End Date

Rent Management

The system shall:

- Record rent payment
- Calculate outstanding balance
- Calculate advance payments
- Calculate partial payments

- Track payment history

Receipt Module

Automatically generate:

- Receipt Number
- Tenant Name
- Shop Number
- Amount Paid
- Balance
- Date
- Collector Name

Allow

- Print
- Download PDF

Notification Module

Automatically send

SMS

Email

Future:

WhatsApp

Messages include

Payment confirmation

Rent reminders

Lease expiry reminders

Maintenance updates

Reports Module

Generate

Daily revenue

Monthly revenue

Annual revenue

Outstanding balances

Occupancy report

Vacant shops

Tenant payment history

Collector performance

Maintenance Module

Tenants can

Submit maintenance request

Manager can

Assign technician

Update status

Statuses

Pending

In Progress

Completed

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7. Non-functional Requirements

Security

- Firebase Authentication
- Role-based permissions

Performance

- Load pages within 3 seconds

Availability

- 99% uptime

Reliability

- Automatic backups

Usability

- Mobile responsive

Scalability

- Support multiple buildings

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8. Business Rules

A tenant cannot occupy more than one shop unless approved.

Each shop belongs to one floor.

Each floor belongs to one building.

Every payment must generate one receipt.

Receipt numbers must be unique.

Only administrators can delete records.

Owners cannot modify payments after approval.

Late payments shall automatically appear in arrears reports.

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9. User Roles

Administrator

Full system access

Building Owner

Read-only financial dashboard

Property Manager

Manage operations

Tenant

View personal information only

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10. Database Collections

Users

Buildings

Floors

Shops

Tenants

Leases

Payments

Receipts

Maintenance Requests

Notifications

Audit Logs

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11. Dashboard Features

Owner Dashboard

- Total Revenue
- Occupied Shops
- Vacant Shops
- Outstanding Rent
- Recent Payments
- Revenue Graph

Property Manager Dashboard

- Today's Collections
- Tenants Due
- New Maintenance Requests

Tenant Dashboard

- Balance
- Payment History
- Receipts
- Lease Information

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12. Enterprise Architecture Mapping

Business Layer

- Property Management
- Finance
- Administration
- Customer Service

Application Layer

- Tenant Module
- Rent Module
- Receipt Module
- Notification Module

- Reports Module

Data Layer

- Firebase Firestore

Technology Layer

- React
- Tailwind CSS
- Node.js
- Express.js
- Firebase

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13. Success Criteria

The project will be considered successful if:

- All rent payments are stored digitally.
- Receipts are generated automatically.
- Payment history can be retrieved at any time.
- Owners can monitor collections in real time.
- Reports are generated instantly.
- The system reduces disputes caused by missing records.
- Different user roles can securely access only the information relevant to them.

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14. Future Enhancements

- MTN Mobile Money Integration
- Airtel Money Integration
- QR Code Receipt Verification
- Barcode Shop Scanning
- AI-Based Rent Prediction
- Multi-language Support
- Mobile Application (Android & iOS)
- Digital Lease Signing
- Automatic Rent Escalation Based on Lease Terms

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15. Conclusion

RentiHub is an Enterprise Property Management System designed to modernize rent collection and property administration in Uganda. By replacing manual record books with a centralized digital platform, the system improves transparency, accountability, efficiency, and decision-making for property owners, managers, and tenants.

The architecture is scalable, secure, and aligned with Enterprise Architecture principles by integrating business processes, applications, data, and technology into a unified solution.